Question: What are examples of measures (so that we don't speak about $\emptyset$)?

Example 2.26 (Measures)

1) If $(\Omega, \mathcal{F})$ is a measurable space, then $\mu(A) = 0$, $A \in \mathcal{F}$, and $\mu(A) = \infty \mathbb{1}_{\{A \neq \emptyset\}}$ (with the convention $\infty \cdot 0 = 0$) are *trivial measures*. They are valid measures even if $\mathcal{F} = \mathcal{P}(\Omega)$.

2) For uncountable Ω, consider the countable-cocountable σ-algebra $\mathcal{F} = \{A \subseteq \Omega : A \text{ or } A^c \text{ is countable}\}$. Then $\mu(A) = \mathbb{1}_{\{A \text{ uncountable}\}}$ is a measure on $\mathcal{F}$:

   i) $\mu : \mathcal{F} \to [0,1] \subseteq [0, \infty]$ ✓;

   ii) $\mu(\emptyset) \underset{\emptyset \text{ count.}}{=} 0$ ✓; and

   iii) Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F}$, $A_i \cap A_j = \emptyset \; \forall i \neq j$. Then

   Case 1: A_i countable $\forall i \in \mathbb{N} \Rightarrow \biguplus_{i=1}^\infty A_i$ countable $\Rightarrow \mu(\biguplus_{i=1}^\infty A_i) = 0$. Also, $\mu(A_i) \underset{A_i \text{ count.}}{=} 0$ for all i, so indeed $\sum_{i=1}^\infty \mu(A_i) = 0$.

   Case 2: $\exists A_k : A_k$ is uncountable. Then $\biguplus_{i=1}^\infty A_i$ is uncountable and thus $\mu(\biguplus_{i=1}^\infty A_i) = 1$. Also, A_k^c is countable and $A_i \underset{A_i \cap A_k = \emptyset}{\subseteq} A_k^c$ $\forall i \neq k$, so A_i must be countable $\forall i \neq k \Rightarrow \mu(A_i) = 0 \; \forall i \neq k$ and $\mu(A_k) = 1$, so indeed $\sum_{i=1}^\infty \mu(A_i) = 1$.

3) For countable Ω consider $\mathcal{F} = \mathcal{P}(\Omega)$. Then $\forall\, f : \Omega \to [0, \infty]$,

$$\mu(A) := \sum_{w \in A} f(w), \quad A \in \mathcal{F},$$

defines a measure on $\mathcal{F}$:

i) $f \geq 0 \Rightarrow \mu : \mathcal{F} \to [0, \infty]$.

ii) $\mu(\emptyset) = \sum_{\omega \in \emptyset} f(\omega) \overset{\text{empty}}{\underset{\text{sum}}{=}} 0$.

iii) If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F}$, $A_i \cap A_j = \emptyset \;\forall\, i \neq j$, then

$$\mu\left(\biguplus_{i=1}^{\infty} A_i\right) \underset{\text{def.}}{=} \sum_{\omega \in \biguplus_{i=1}^{\infty} A_i} f(\omega) = \sum_{i=1}^{\infty} \sum_{\omega \in A_i} f(\omega) \underset{\text{def.}}{=} \sum_{i=1}^{\infty} \mu(A_i).$$

- If $f \equiv 1$, then $\mu(A) = |A|$ is the *counting measure*.
- If, for $\tilde{\omega} \in \Omega$, $f(w) = \mathbb{1}_{\{\omega = \tilde{\omega}\}}$, then $\mu(A) = \mathbb{1}_{\{\tilde{\omega} \in A\}}$ is the *Dirac measure* or *point mass* or *unit mass* of $\tilde{\omega}$.

Question: What properties do measures have? Some even extend to (semi)rings.

Proposition 2.27 (Basic properties of measures)

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space.

1) $A, B \in \mathcal{F} \Rightarrow \mu(A \cup B) + \mu(A \cap B) = \mu(A) + \mu(B)$. If μ is finite, then $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B)$.
2) $A, B \in \mathcal{F}$, $A \subseteq B \Rightarrow \mu(A) \leq \mu(B)$ (*monotonicity*).
   If $\mu(A) < \infty$, then $\mu(B \backslash A) = \mu(B) - \mu(A)$ (*subtractivity*).
3) $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F} \Rightarrow \mu(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$ (*σ-subadditivity*).
4) If μ is finite, $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F}$, $S_{j,n} := \sum_{J \subseteq \{1,\dots,n\} : |J| = j} \mu(\bigcap_{k \in J} A_k)$, then
$$\mu\left(\bigcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{n} (-1)^{j-1} S_{j,n} \quad (\textit{inclusion–exclusion principle}).$$
5) If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F}$, $A_i \nearrow$, then $\mu(\bigcup_{i=1}^{\infty} A_i) = \lim_{n \to \infty} \mu(A_n)$ (*cont. from below*).
6) If $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F}$, $A_i \searrow$ and $\mu(A_1) < \infty$, then $\mu(\bigcap_{i=1}^{\infty} A_i) = \lim_{n \to \infty} \mu(A_n)$ (*continuity from above*).
7) If $\{B_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F}$ forms a partition of Ω, $A \in \mathcal{F}$, then $\mu(A) = \sum_{i=1}^{\infty} \mu(A \cap B_i)$ (*law of total measure*).

Proof.

1) $A \cup B = A \uplus (B \backslash A)$ and $B = (A \cap B) \uplus (B \backslash A)$. By additivity, $\mu(A \cup B) \underset{(*)}{=} \mu(A) + \mu(B \backslash A)$ and $\mu(B) \underset{(**)}{=} \mu(A \cap B) + \mu(B \backslash A)$. Adding opposite sides gives $\mu(A \cup B) + \mu(A \cap B) + \mu(B \backslash A) = \mu(A) + \mu(B \backslash A) + \mu(B)$.
   - If $\mu(B \backslash A) < \infty$, subtract $\mu(B \backslash A)$ and we are done.
   - And if $\mu(B \backslash A) = \infty$, then $\mu(A \cup B) \underset{(*)}{=} \infty$ and $\mu(B) \underset{(**)}{=} \infty$, so the formula as stated is still valid (it then states "$\infty = \infty$").

   If μ is finite, subtract $\mu(A \cap B)$ from both sides in the just shown first statement.
2) $\mu(B) \underset{(**)}{=} \mu(A \cap B) + \mu(B \backslash A) \overset{A \cap B \underset{\text{ass.}}{=} A}{\underset{(***)}{=}} \mu(A) + \mu(B \backslash A) \geq \mu(A)$. If $\mu(A) < \infty$, subtract it to obtain $\mu(B \backslash A) \underset{(***)}{=} \mu(B) - \mu(A)$ (irresp. of the value of $\mu(B)$).
3) Let $B_1 := A_1$, $B_n := A_n \backslash \bigcup_{i=1}^{n-1} A_i$, $n \geq 2 \Rightarrow B_n$'s are pairwise disjoint $\Rightarrow \bigcup_{i=1}^{\infty} A_i = \lim_{N \to \infty} \bigcup_{i=1}^{N} A_i = \lim_{N \to \infty} \biguplus_{i=1}^{N} B_i = \biguplus_{i=1}^{\infty} B_i \Rightarrow \mu(\bigcup_{i=1}^{\infty} A_i) = \mu(\biguplus_{i=1}^{\infty} B_i) \underset{\sigma\text{-add.}}{=} \sum_{i=1}^{\infty} \mu(B_i) \overset{B_i \subseteq A_i}{\underset{\text{mon.}}{\leq}} \sum_{i=1}^{\infty} \mu(A_i)$.
4) Induction in n based on 1):

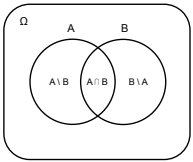
   $n = 2 : \mu(A_1 \cup A_2) \underset{1)}{=} \mu(A_1) + \mu(A_2) - \mu(A_1 \cap A_2) = \sum_{j=1}^{2} (-1)^{j-1} S_{j,2}$.

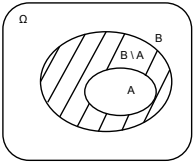
   $n \Rightarrow n + 1$:

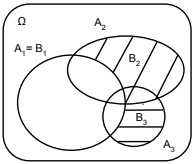
$$
\begin{aligned}
\mu\Big(\bigcup_{j=1}^{n+1} A_j\Big) &= \mu\Big(\Big(\bigcup_{j=1}^{n} A_j\Big) \cup A_{n+1}\Big) \\
&\underset{1)}{=} \mu\Big(\bigcup_{j=1}^{n} A_j\Big) + \mu(A_{n+1}) - \mu\Big(\Big(\bigcup_{j=1}^{n} A_j\Big) \cap A_{n+1}\Big) \\
&\overset{\text{hypo.}}{\underset{\text{distr.}}{=}} \sum_{j=1}^{n} (-1)^{j-1} S_{j,n} + \mu(A_{n+1}) - \mu\Big(\bigcup_{j=1}^{n} (A_j \cap A_{n+1})\Big) \\
&\underset{\text{hypo.}}{=} \sum_{j=2}^{n} (-1)^{j-1} S_{j,n} + \underbrace{S_{1,n}}_{\underset{\text{def.}}{=} \sum_{i=1}^{n} \mu(A_i)} + \mu(A_{n+1}) - \sum_{j=1}^{n} (-1)^{j-1} \sum_{J \subseteq \{1,\dots,n\} : |J| = j} \mu\Big(\Big(\bigcap_{k \in J} A_k\Big) \cap A_{n+1}\Big) \\
&= \sum_{j=2}^{n} (-1)^{j-1} S_{j,n} + S_{1,n+1} - \sum_{j=1}^{n} (-1)^{j-1} \sum_{J \subseteq \{1,\dots,n\} : |J| = j} \mu\Big(\Big(\bigcap_{k \in J} A_k\Big) \cap A_{n+1}\Big) \\
&\underset{(*)}{=} S_{1,n+1} + \sum_{j=2}^{n+1} (-1)^{j-1} S_{j,n+1} = \sum_{j=1}^{n+1} (-1)^{j-1} S_{j,n+1},
\end{aligned}
$$

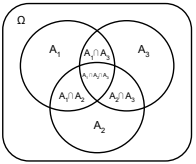
where $(*)$ holds since the first sum contains all intersections of at least two sets of which none contains A_{n+1}, and the last sum contains all intersections of at least two sets where one is A_{n+1}, so together we obtain the sum containing all intersections of at least two sets among $A_1, \dots, A_{n+1}$.

Reminder: If possible and helpful, draw Venn diagrams. For 1)–4):









5) $A_0 := \emptyset$, $A_i \nearrow \Rightarrow \mu(\bigcup_{i=1}^\infty A_i) \overset{\text{disjoint}}{\underset{\text{partition}}{=}} \mu(\biguplus_{i=1}^\infty (A_i \backslash A_{i-1})) \underset{\sigma\text{-add.}}{=} \sum_{i=1}^\infty \mu(A_i \backslash A_{i-1}) \underset{\text{def.}}{=}$
$\lim_{n\to\infty} \sum_{i=1}^n \mu(A_i \backslash A_{i-1}) \underset{\text{add.}}{=} \lim_{n\to\infty} \mu(\biguplus_{i=1}^n (A_i \backslash A_{i-1})) \underset{A_i \nearrow}{=} \lim_{n\to\infty} \mu(A_n)$.

6) Let $B_i := A_1 \backslash A_i = A_1 \cap A_i^c$, $i \in \mathbb{N}$. Then $B_i \nearrow \bigcup_{i=1}^\infty B_i = \bigcup_{i=1}^\infty (A_1 \cap A_i^c) \underset{\text{distr.}}{=}$
$A_1 \cap \bigcup_{i=1}^\infty A_i^c \underset{\text{De Morgan}}{=} A_1 \cap (\bigcap_{i=1}^\infty A_i)^c = A_1 \backslash \bigcap_{i=1}^\infty A_i$. Since $\mu(A_1) < \infty$,

$$\mu(A_1) - \mu\left(\bigcap_{i=1}^\infty A_i\right) \underset{2)}{=} \mu\left(A_1 \backslash \bigcap_{i=1}^\infty A_i\right) = \mu\left(\bigcup_{i=1}^\infty B_i\right) \overset{B_i \nearrow}{\underset{5)}{=}} \lim_{n\to\infty} \mu(B_n)$$

$$\underset{\text{def.}}{=} \lim_{n\to\infty} \mu(A_1 \backslash A_n) \overset{2)}{\underset{\mu(A_n) < \infty}{=}} \lim_{n\to\infty} (\mu(A_1) - \mu(A_n)) = \mu(A_1) - \lim_{n\to\infty} \mu(A_n).$$

Since $\mu(A_1) < \infty$, subtract $\mu(A_1)$ from both sides to get the result.

7) $\mu(A) = \mu(A \cap \Omega) \underset{\text{part.}}{=} \mu(A \cap \biguplus_{i=1}^\infty B_i) \underset{\text{distr.}}{=} \mu(\biguplus_{i=1}^\infty (A \cap B_i)) \underset{\sigma\text{-add.}}{=} \sum_{i=1}^\infty \mu(A \cap B_i)$. $\square$

Proposition 2.28 (Uniqueness)
Let μ, ν be measures on $(\Omega, \mathcal{F})$ and $\mathcal{A}$ a π-system such that $\exists\,(A_i)_{i\in\mathbb{N}} \subseteq \mathcal{A}$ with $\bigcup_{i=1}^{\infty} A_i = \Omega$, $\mu(A_i) < \infty\ \forall\, i \in \mathbb{N}$. If $\mu|_{\mathcal{A}} = \nu|_{\mathcal{A}}$, then $\mu|_{\sigma(\mathcal{A})} = \nu|_{\sigma(\mathcal{A})}$. In particular, if $\sigma(\mathcal{A}) = \mathcal{F}$, then $\mu = \nu$.

Proof.

1) For $B \in \mathcal{A} : \mu(B) < \infty$, let $\mathcal{D}_B := \{A \in \sigma(\mathcal{A}) : \mu(A \cap B) = \nu(A \cap B)\}$. We first show that $\mathcal{D}_B$ is a Dynkin system:
   i) $\Omega \in \sigma(\mathcal{A}) : \mu(\Omega \cap B) = \mu(B) \underset{B\,\in\,\mathcal{A}}{=} \nu(B) = \nu(\Omega \cap B) \Rightarrow \Omega \in \mathcal{D}_B$;
   ii) $A \in \mathcal{D}_B \Rightarrow A \in \sigma(\mathcal{A}) \Rightarrow A^c \in \sigma(\mathcal{A})$. Furthermore, $\mu(A^c \cap B) + \mu(A \cap B) \underset{\sigma\text{-add.}}{=} \mu((A^c \cap B) \uplus (A \cap B)) \overset{\text{tot.}}{\underset{\text{meas.}}{=}} \mu(B) \underset{B\,\in\,\mathcal{A}}{=} \nu(B) = \ldots \overset{\text{same}}{\underset{\text{backwards}}{=}} \nu(A^c \cap B) + \nu(A \cap B)$, with $\mu(A \cap B) \underset{A\,\in\,\mathcal{D}_B}{=} \nu(A \cap B) \overset{\mu(A\cap B) \le \mu(B) < \infty}{\underset{\text{subtract}}{\Rightarrow}} \mu(A^c \cap B) = \nu(A^c \cap B) \Rightarrow A^c \in \mathcal{D}_B$; and
   iii) $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{D}_B : A_i \cap A_j = \emptyset\ \forall\, i \neq j \Rightarrow \mu((\biguplus_{i=1}^{\infty} A_i) \cap B) \underset{\text{distr.}}{=} \mu(\biguplus_{i=1}^{\infty}(A_i \cap B)) \underset{\sigma\text{-add.}}{=} \sum_{i=1}^{\infty} \mu(A_i \cap B) \underset{\text{def. }\mathcal{D}_B}{=} \sum_{i=1}^{\infty} \nu(A_i \cap B) = \ldots \overset{\text{same}}{\underset{\text{backwards}}{=}} \nu((\biguplus_{i=1}^{\infty} A_i) \cap B) \Rightarrow \biguplus_{i=1}^{\infty} A_i \in \mathcal{D}_B$.
2) By ass., $\mathcal{A}$ is a π-system, so $\forall\, A_1, A_2 \in \mathcal{A}$, $A_1 \cap A_2 \in \mathcal{A}$ and thus $\mu(A_1 \cap A_2) \overset{\mu\,\equiv\,\nu}{\underset{\text{on }\mathcal{A}}{=}} \nu(A_1 \cap A_2)$, so $\mathcal{A} \subseteq \mathcal{D}_B$. By Dynkin's π-λ T., $\sigma(\mathcal{A}) \subseteq \mathcal{D}_B \underset{\text{def. }\mathcal{D}_B}{\subseteq} \sigma(\mathcal{A})$.

So $\mathcal{D}_B = \sigma(\mathcal{A})\ \forall\, B \in \mathcal{A} : \mu(B) < \infty$, thus $\mu(A \cap B) = \nu(A \cap B)\ \forall\, A \in \sigma(\mathcal{A})$, $\forall\, B \in \mathcal{A} : \mu(B) < \infty$.

3) i) By ass., $\exists\, (A_i)_{i\in\mathbb{N}} \underset{\text{ass.}}{\subseteq} \mathcal{A} : \bigcup_{i=1}^{\infty} A_i = \Omega$, $\mu(A_i) < \infty\ \forall\, i \in \mathbb{N}$. Let $A'_0 := \emptyset$ and $A'_n := \bigcup_{i=1}^{n} A_i$, $n \in \mathbb{N}$. Then $A'_n \nearrow \Omega$ and $A'_n = \biguplus_{i=1}^{n} (A_i \setminus A'_{i-1})$ is a disjoint decomposition of A'_n into sets of $\sigma(\mathcal{A})$.

ii) $\forall\, A \in \sigma(\mathcal{A})$,

$$\mu(A \cap A'_n) \underset{\text{distr.}}{=} \mu\left(\biguplus_{i=1}^{n} A \cap A_i \cap A'^c_{i-1}\right) \underset{\sigma\text{-add.}}{=} \sum_{i=1}^{n} \mu(\underbrace{A \cap A'^c_{i-1}}_{\in\sigma(\mathcal{A})} \cap \underbrace{A_i}_{\in\mathcal{A}})$$

$$\overset{\mu(A_i)<\infty}{\underset{2)}{=}} \sum_{i=1}^{n} \nu(\underbrace{A \cap A'^c_{i-1}}_{\in\sigma(\mathcal{A})} \cap \underbrace{A_i}_{\in\mathcal{A}}) = \ldots \overset{\text{same}}{\underset{\text{backwards}}{=}} \nu(A \cap A'_n).$$

iii) Therefore, $\forall\, A \in \sigma(\mathcal{A})$, $\mu(A) = \mu(A \cap \Omega) \overset{\text{i)}}{\underset{\text{cont. below}}{=}} \lim_{n\to\infty} \mu(A \cap A'_n) \underset{\text{ii)}}{=} \lim_{n\to\infty} \nu(A \cap A'_n) = \ldots \overset{\text{same}}{\underset{\text{backwards}}{=}} \nu(A)$. $\square$

By i), σ-finiteness on $\mathcal{A}$ can be replaced by the existence of $(A_i)_{i\in\mathbb{N}} \subseteq \mathcal{A}$, $A_i \nearrow \Omega$ with $\mu(A_i) < \infty$, $i \in \mathbb{N}$, (*exhausting sequence*). This trivially holds if μ is finite.

Example 2.29 (Product measure)

For $j = 1, \dots, d$, let Ω_j be equipped with a σ-algebra $\mathcal{F}_j$. The product space $\prod_{j=1}^d \Omega_j$ can then be equipped with the product-σ-algebra $\bigotimes_{j=1}^d \mathcal{F}_j \underset{\text{P. 2.16}}{=} \sigma(\prod_{j=1}^d A_j : A_j \in \mathcal{F}_j \; \forall\, j)$. If μ_j is a σ-finite measure on $(\Omega_j, \mathcal{F}_j) \; \forall\, j$, then

$$\left(\prod_{j=1}^d \mu_j\right)\left(\prod_{j=1}^d A_j\right) := \prod_{j=1}^d \mu_j(A_j), \quad \prod_{j=1}^d A_j \in \bigotimes_{j=1}^d \mathcal{F}_j,$$

is the *product measure* on $(\prod_{j=1}^d \Omega_j, \bigotimes_{j=1}^d \mathcal{F}_j)$; by P. 2.28, it suffices to define $\prod_{j=1}^d \mu_j$ on the π-system $\mathcal{A} = \{\prod_{j=1}^d A_j : A_j \in \mathcal{F}_j \; \forall\, j\}$.

The product measure is indeed a measure since:

i) $\mu_j : \mathcal{F}_j \to [0, \infty]$, $j = 1, \dots, d \Rightarrow \prod_{j=1}^d \mu_j : \bigotimes_{j=1}^d \mathcal{F}_j \to [0, \infty]$.

ii) In $\prod_{j=1}^d \Omega_j$, $\prod_{j=1}^d A_j = \emptyset$ iff $\exists\, k \in \{1, \dots, d\} : A_k = \emptyset$. As such, if $A_k = \emptyset$ for at least one k, then

$$\left(\prod_{j=1}^d \mu_j\right)(\emptyset) = \left(\prod_{j=1}^d \mu_j\right)\left(\prod_{j=1}^d A_j\right) = \prod_{j=1}^d \mu_j(A_j) \underset{\mu_k(A_k) = 0}{=} 0.$$

iii) If $\{A_i\}_{i\in\mathbb{N}} \subseteq \bigotimes_{j=1}^d \mathcal{F}_j$, $A_i \cap A_j = \emptyset \; \forall\, i \neq j$, then $A_i = \prod_{j=1}^d A_{i,j}$ for $A_{i,j} \in \mathcal{F}_j$ $\forall\, i, j$. With

$$\biguplus_{i=1}^{\infty} A_i = \biguplus_{i=1}^{\infty} \prod_{j=1}^{d} A_{i,j} \underset{(*)}{=} \prod_{j=1}^{d} \biguplus_{i=1}^{\infty} A_{i,j}$$

we have

$$\begin{aligned}
\left(\prod_{j=1}^{d} \mu_j\right)\left(\biguplus_{i=1}^{\infty} A_i\right) &\underset{(*)}{=} \left(\prod_{j=1}^{d} \mu_j\right)\left(\prod_{j=1}^{d} \biguplus_{i=1}^{\infty} A_{i,j}\right) \underset{\text{def.}}{=} \prod_{j=1}^{d} \mu_j\left(\biguplus_{i=1}^{\infty} A_{i,j}\right) \\
&\underset{\sigma\text{-add.}}{\overset{\mu_j}{=}} \prod_{j=1}^{d} \sum_{i=1}^{\infty} \mu_j(A_{i,j}) \underset{\text{terms} \geq 0}{\overset{\text{multiply out}}{=}} \sum_{i=1}^{\infty} \prod_{j=1}^{d} \mu_j(A_{i,j}) \\
&\underset{\text{def.}}{=} \sum_{i=1}^{\infty}\left(\prod_{j=1}^{d} \mu_j\right)\left(\prod_{j=1}^{d} A_{i,j}\right) \underset{\text{def. } A_i}{=} \sum_{i=1}^{\infty}\left(\prod_{j=1}^{d} \mu_j\right)(A_i).
\end{aligned}$$

2.4 Null sets

Definition 2.30 (Null set, a.e., a.s., completeness)
If $(\Omega, \mathcal{F}, \mu)$ is a measure space, every $N \in \mathcal{F} : \mu(N) = 0$ is a *(μ-)null set*. If a statement holds $\forall\, \omega \in \Omega \backslash N$ for a null set N, it holds *(μ-)almost everywhere (a.e.)*, or, if μ is a probability measure, the statement holds *(μ-)almost surely (a.s.)*. If $\mathcal{F}$ contains all subsets of null sets, μ is a *complete* measure.

If a statement holds $\forall\, \omega \in \Omega$, one says it holds "everywhere", "surely" (or "point-wise"). Whether a statement holds everywhere/surely or only a.e./a.s. typically does not matter as every (in)equality involving measures holds irrespectively of changes on null sets.

Question: What collection of null sets is still a null set?

Lemma 2.31 (Countable union of null sets)
A countable union of null sets in $\mathcal{F}$ is a null set in $\mathcal{F}$.

Proof. If $\{N_i\}_{i \in \mathbb{N}} \subseteq \mathcal{F}$ are null sets, then $\bigcup_{i=1}^{\infty} N_i \in \mathcal{F}$ and $0 \leq \mu(\bigcup_{i=1}^{\infty} N_i) \underset{\sigma\text{-subadd.}}{\leq} \sum_{i=1}^{\infty} \mu(N_i) = \sum_{i=1}^{\infty} 0 = 0.$ $\square$

Question: If μ is not complete, can we extend it to a complete measure $\bar{\mu}$ so that all subsets N' of null sets N are measurable sets, with $\bar{\mu}(N') = 0$?

Theorem 2.32 (Completion of a σ-algebra and measure)
Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and $\mathcal{N} = \{N \in \mathcal{F} : \mu(N) = 0\}$. Then

1) $\bar{\mathcal{F}} := \{A \cup N' : A \in \mathcal{F},\ N' \subseteq N \text{ for } N \in \mathcal{N}\}$ is a σ-algebra on Ω, the *completion* of $\mathcal{F}$.
2) $\bar{\mu}(A \cup N') := \mu(A)\ \forall A \in \mathcal{F},\ \forall N' \subseteq N$ for $N \in \mathcal{N}$ uniquely extends μ to a complete measure on $\bar{\mathcal{F}}$.

Proof.

1) i) $\Omega = \Omega \cup \emptyset \in \bar{\mathcal{F}}$;

ii) $\bar{A} \in \bar{\mathcal{F}} \Rightarrow \bar{A} = A \cup N'$ for some $A \in \mathcal{F}$, $N \in \mathcal{N} : N \supseteq N'$. Wlog, assume $A \cap N = \emptyset$; otherwise consider $N' \leftarrow N' \backslash A$ and $N \leftarrow N \backslash A \in \mathcal{N}$. Then $\bar{A} = A \cup N' = A \cup \emptyset \cup \emptyset \cup N' \underset{N' \subseteq N}{\overset{A \cap N = \emptyset}{=}} (A \cap N^c) \cup (A \cap N') \cup (N \cap N^c) \cup (N \cap N') \underset{\text{distr.}}{=} (A \cup N) \cap (N^c \cup N')$, so that $\bar{A}^c = ((A \cup N) \cap (N^c \cup N'))^c \underset{\text{De Morgan}}{=} \underbrace{(A \cup N)^c}_{\in \mathcal{F} \text{ as } A, N \in \mathcal{F}} \cup \underbrace{(N \backslash N')}_{\subseteq N \in \mathcal{N}} \in \bar{\mathcal{F}}$.

iii) $\{\bar{A}_i\}_{i\in\mathbb{N}} \subseteq \bar{F} \Rightarrow \bar{A}_i = A_i \cup N_i'$ for $A_i \in \mathcal{F}$, $N_i \in \mathcal{N} : N_i \supseteq N_i' \; \forall\, i \in \mathbb{N} \Rightarrow \bigcup_{i=1}^\infty \bar{A}_i = \bigcup_{i=1}^\infty (A_i \cup N_i') = \underbrace{(\bigcup_{i=1}^\infty A_i)}_{\in \mathcal{F}} \cup \underbrace{(\bigcup_{i=1}^\infty N_i')}_{\subseteq \bigcup_{i=1}^\infty N_i \underset{N_i \in \mathcal{F}}{\in} \mathcal{N}} \in \bar{\mathcal{F}}$.

2)
- $\bar{\mu}$ is well-defined on $\bar{\mathcal{F}}$ since for $A_1 \cup N_1' = A_2 \cup N_2'$, we have $A_1 \subseteq A_1 \cup N_1' = A_2 \cup N_2'$ and so $\mu(A_1) \leq \mu(A_2) + 0$, and likewise $\mu(A_2) \leq \mu(A_1)$, so $\mu(A_1) = \mu(A_2)$ and thus $\bar{\mu}(A_1) = \mu(A_1) = \mu(A_2) = \bar{\mu}(A_2)$.
- $\bar{\mu}$ is a measure on $\bar{F}$ (by definition, we already know that $\bar{\mu}|_{\mathcal{F}} = \mu$):
  i) $\mu : \mathcal{F} \to [0,\infty] \Rightarrow \bar{\mu} : \bar{\mathcal{F}} \to [0,\infty]$;
  ii) $\emptyset = \emptyset \cup \emptyset \in \bar{\mathcal{F}} \Rightarrow \bar{\mu}(\emptyset) = \bar{\mu}(\emptyset \cup \emptyset) = \mu(\emptyset) = 0$; and
  iii) $\forall\, i \in \mathbb{N}$, let $\bar{A}_i = A_i \cup N_i'$ for $A_i \in \mathcal{F}$, $N_i \in \mathcal{N} : N_i \supseteq N_i'$ and $\bar{A}_i \cap \bar{A}_j = \emptyset \; \forall\, i \neq j$. Then $\bar{\mu}(\biguplus_{i=1}^\infty \bar{A}_i) \underset{\text{def. } \bar{A}_i}{=} \bar{\mu}((\biguplus_{i=1}^\infty A_i) \cup (\biguplus_{i=1}^\infty N_i')) \underset{(*)}{=} \mu(\biguplus_{i=1}^\infty A_i) \underset{\sigma\text{-add.}}{=} \sum_{i=1}^\infty \mu(A_i) \underset{\text{def. } \bar{\mu}}{=} \sum_{i=1}^\infty \bar{\mu}(A_i \cup N_i') \underset{\text{def. } \bar{A}_i}{=} \sum_{i=1}^\infty \bar{\mu}(\bar{A}_i)$, where $(*)$ holds since $\biguplus_{i=1}^\infty A_i \in \mathcal{F}$, $\biguplus_{i=1}^\infty N_i' \subseteq \bigcup_{i=1}^\infty N_i \underset{\text{L. 2.31}}{\in} \mathcal{N}$.
- Uniqueness: Suppose $\exists$ a complete measure $\bar{\nu}$ on $\bar{\mathcal{F}} : \bar{\nu}(A \cup N') = \mu(A)$ $\forall\, A \in \mathcal{F}$, $\forall\, N' \subseteq N$, $\forall\, N \in \mathcal{N}$. Then $\bar{\nu}(A \cup N') \underset{\text{subadd.}}{\leq} \bar{\nu}(A) + \bar{\nu}(N') = \bar{\nu}(A) + 0 \underset{\bar{\nu} \text{ extends } \mu}{=} \mu(A) \underset{\bar{\mu} \text{ extends } \mu}{=} \bar{\mu}(A) \underset{\text{mon.}}{\leq} \bar{\mu}(A \cup N')$. Likewise $\bar{\mu}(A \cup N') \leq \bar{\nu}(A \cup N')$, so $\bar{\mu}(A \cup N') = \bar{\nu}(A \cup N')$. $\square$

2.5 Construction of measures

Question: How can we construct measures on general measurable spaces?

- **Main idea:** Start from a 'premeasure' μ_0 (a notion of measure) on a sufficiently simple system of sets $\mathcal{A} \subseteq \mathcal{P}(\Omega)$ and extend μ_0 via an 'outer measure' μ^* to a measure μ on $\sigma(\mathcal{A})$. Results of this form are referred to as *Carathéodory extension theorem*, attributed to Constantin Carathéodory (1873–1950).
- To be consistent (not leading to contradictions), $\mathcal{A}$ should have some structure:
  - Folland (1999, T. 1.13, T. 1.14), Durrett (2019, T. A.1.1, T. A.1.3): $\mathcal{A}$ is an algebra ($\Rightarrow$ Hahn–Kolmogorov theorem)
  - Bauer (2001, T. 5.1, T. 5.3, T. 5.6), Wikipedia: $\mathcal{A}$ is a ring ($\Rightarrow$ Carathéodory's extension theorem)
  - Klenke (2008, T. 1.53), Schilling (2006, T. 6.1): $\mathcal{A}$ is a semiring ($\Rightarrow$ Carathéodory's extension theorem; most general: Klenke)
- The more structure (from semirings to rings to algebras) the easier it is to prove the extension theorem, but typically the harder it is to apply it for a specific construction as more properties need to be verified (but note that one also has less properties available to work with). We consider semirings.

Premeasures have the properties of measures but are defined on a smaller domain.

Definition 2.33 (Premeasure, σ-finite)
Let $\mathcal{A}$ be a semiring on Ω. A *premeasure* μ_0 on $\mathcal{A}$ satisfies

i) $\mu_0 : \mathcal{A} \to [0, \infty]$;

ii) $\mu_0(\emptyset) = 0$; and

iii) $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{A}$, $A_i \cap A_j = \emptyset \ \forall\, i \neq j$, and $\biguplus_{i=1}^{\infty} A_i \in \mathcal{A} \Rightarrow \mu_0(\biguplus_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu_0(A_i)$ (*σ-additivity*).

If $\Omega = \bigcup_{i=1}^{\infty} A_i$ for $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{A} : \mu_0(A_i) < \infty \ \forall\, i$, then μ_0 is *σ-finite*.

Outer measures are used to approximate volumes from 'above' (the 'outside').

Definition 2.34 (Outer measure)
An *outer measure* $\mu^* : \mathcal{P}(\Omega) \to [0, \infty]$ satisfies

i) $\mu^*(\emptyset) = 0$;

ii) $A, B \subseteq \Omega : A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B)$ (monotonicity); and

iii) $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{P}(\Omega) \Rightarrow \mu^*(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$ (σ-subadditivity).

Steps of the main idea: Start with a premeasure μ_0 on a semiring $\mathcal{A}$.

1) Show that the premeasure μ_0 defined on the semiring $\mathcal{A}$ induces an outer measure μ^*.
2) Show that μ^* extends μ_0 to $\mathcal{P}(\Omega)$, i.e. $\mu^*|_{\mathcal{A}} = \mu_0|_{\mathcal{A}}$.
3) Show that $\mathcal{A} \subseteq \mathcal{A}^*$ where $\mathcal{A}^*$ is the family of *Carathéodory-measurable sets*

$$\mathcal{A}^* := \{A \subseteq \Omega : \mu^*(B) = \mu^*(B \cap A) + \mu^*(B \cap A^c)\ \forall\, B \subseteq \Omega\}.$$

4) Show that $\mathcal{A}^*$ is a σ-algebra on Ω ($\underset{3)}{\Rightarrow} \sigma(\mathcal{A}) \subseteq \mathcal{A}^*$) and μ^* is a measure on $\mathcal{A}^*$.
5) Then $\mu := \mu^*|_{\sigma(\mathcal{A})}$ is a measure. Show that $\mu|_{\mathcal{A}} = \mu_0|_{\mathcal{A}}$, i.e. μ extends μ_0 to $\sigma(\mathcal{A})$.
6) Show that if μ_0 is σ-finite on $\mathcal{A}$, the extension μ of μ_0 to $\sigma(\mathcal{A})$ is unique.

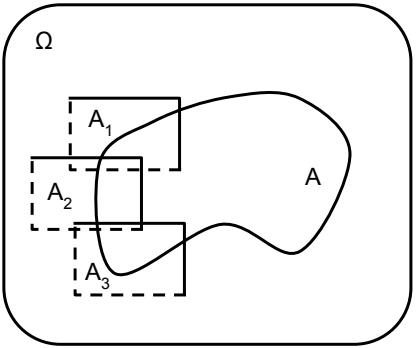
Theorem 2.35 (Carathéodory extension theorem)
Let $\mathcal{A}$ be a semiring on Ω and μ_0 a σ-finite premeasure on $\mathcal{A}$. Then μ_0 has a unique extension to a σ-finite measure μ on $\sigma(\mathcal{A})$.

Proof. Step 1) is instructive. Step 2) is more work (partly because we start with semirings). Steps 3)–5) are the main parts of the theorem. Step 6) is uniqueness.

1) Let $\mu^* : \mathcal{P}(\Omega) \to [0, \infty]$ be defined by

$$\mu^*(A) := \inf_{\substack{A \subseteq \bigcup_{i=1}^\infty A_i, \\ A_i \in \mathcal{A}}} \sum_{i=1}^\infty \mu_0(A_i),$$

where $\inf \emptyset = \infty$. In particular, if $\nexists$ covering of A by a countable union of sets from $\mathcal{A}$, then $\mu^*(A) = \infty$, and if $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{A}$ such that $\mu_0(A_i) = 0$ $\forall\, i$, then $\mu^*(A) = 0$.



We show that μ^* is an outer measure:

i) $A_i = \emptyset \;\forall\, i \in \mathbb{N} \Rightarrow \mu^*(\emptyset) = 0$;

ii) Let $A, B \subseteq \Omega : A \subseteq B$. If $\bigcup_{i=1}^\infty A_i \supseteq B$ then $\bigcup_{i=1}^\infty A_i \supseteq A \Rightarrow \mu^*(A) \leq \mu^*(B)$ (as the $\inf$ over a larger number of coverings can only be smaller);

iii) Let $\{A_i\}_{i \in \mathbb{N}} \subseteq \mathcal{P}(\Omega)$. Wlog, assume $\mu^*(A_i) < \infty \;\forall\, i \in \mathbb{N}$; otherwise σ-subadditivity trivially holds; "$\infty \underset{\text{ii)}}{\leq} \infty$". By definition of the infimum, $\forall\, \varepsilon > 0$, $\exists$ a covering $\bigcup_{k=1}^\infty A_{i,k} \supseteq A_i$ with $\sum_{k=1}^\infty \mu_0(A_{i,k}) \underset{(*)}{\leq} \mu^*(A_i) + \varepsilon/2^i$, $i \in \mathbb{N}$. Therefore $\bigcup_{i,k=1}^\infty A_{i,k} \supseteq \bigcup_{i=1}^\infty A_i$ is a covering and thus

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \underset{\text{all coverings}}{\overset{\mu^* \text{ smallest over}}{\leq}} \sum_{i,k=1}^{\infty} \mu_0(A_{i,k}) \underset{(*)}{\leq} \sum_{i=1}^{\infty}(\mu^*(A_i) + \varepsilon/2^i) = \varepsilon + \sum_{i=1}^{\infty} \mu^*(A_i).$$

As $\varepsilon > 0$ was arbitrary, we obtain (for $\varepsilon \to 0+$) $\mu^*(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$.

2) To show that $\mu^*|_{\mathcal{A}} = \mu_0|_{\mathcal{A}}$, we proceed in several steps:

2.1) We show that $\mathcal{A}_{\uplus} := \{\biguplus_{i=1}^{n} A_i : A_i \in \mathcal{A},\ n \in \mathbb{N}\}$ is a ring, the *ring generated by $\mathcal{A}$*. To this end, we first cover two auxiliary results:

a) For any $A, B_1, \ldots, B_m \in \mathcal{A}$ there are disjoint $C_1, \ldots, C_k \in \mathcal{A}$: $A \setminus \bigcup_{j=1}^{m} B_j = \biguplus_{l=1}^{k} C_l$. Proof by induction:
$m = 1$: $A, B_1 \in \mathcal{A} \underset{\text{D. 2.3 iii)}}{\Rightarrow} A \setminus B_1$ is a finite disjoint union of sets in $\mathcal{A}$.
$m \Rightarrow m{+}1$: $A \setminus \bigcup_{j=1}^{m+1} B_j = (A \setminus \bigcup_{j=1}^{m} B_j) \setminus B_{m+1} \underset{\text{hypo.}}{=} (\biguplus_{l=1}^{k} C_l) \setminus B_{m+1}$ $\underset{\text{distr.}}{=} \biguplus_{l=1}^{k} (C_l \setminus B_{m+1})$. By D. 2.3 iii), each $C_l \setminus B_{m+1}$ is a finite disjoint union of sets in $\mathcal{A}$, and so is $\biguplus_{l=1}^{k} (C_l \setminus B_{m+1})$ and thus $A \setminus \bigcup_{j=1}^{m+1} B_j$.

b) $\mathcal{A}_{\uplus}$ is a π-system: For $A = \biguplus_{i=1}^{n} A_i \in \mathcal{A}_{\uplus}$ and $B = \biguplus_{j=1}^{m} B_j \in \mathcal{A}_{\uplus}$, we have $A \cap B = \biguplus_{i,j=1}^{n,m} (A_i \cap B_j)$ for $A_i \cap B_j \underset{\text{D. 2.3 ii)}}{\in} \mathcal{A}$, so $A \cap B \in \mathcal{A}_{\uplus}$.

We can now show that $\mathcal{A}_{\uplus}$ is a ring:

i) $\emptyset \in \mathcal{A} \Rightarrow \emptyset \in \mathcal{A}_{\uplus}$

iii) For $A = \biguplus_{i=1}^n A_i \in \mathcal{A}_\uplus$ and $B = \biguplus_{j=1}^m B_j \in \mathcal{A}_\uplus$, we have

$$A \backslash B = \left(\biguplus_{i=1}^n A_i \right) \cap \left(\biguplus_{j=1}^m B_j \right)^c \underset{\text{De Morgan}}{=} \left(\biguplus_{i=1}^n A_i \right) \cap \bigcap_{j=1}^m B_j^c$$

$$\underset{\text{distr.}}{=} \biguplus_{i=1}^n \left(A_i \cap \bigcap_{j=1}^m B_j^c \right) \underset{\text{De Morgan}}{=} \biguplus_{i=1}^n \left(A_i \cap \left(\bigcup_{j=1}^m B_j \right)^c \right)$$

$$= \biguplus_{i=1}^n \left(A_i \backslash \bigcup_{j=1}^m B_j \right) \underset{\text{a)}}{=} \biguplus_{i=1}^n \biguplus_{l=1}^{k_i} C_{i,l} \underset{\text{def.}}{\in} \mathcal{A}_\uplus$$

for some disjoint $C_{i,l} \in \mathcal{A}\ \forall i \in \{1, \dots, n\}, l \in \{1, \dots, k_i\}$.

ii) $A, B \in \mathcal{A}_\uplus \Rightarrow A \cup B = (A \backslash B) \uplus (A \cap B) \uplus (B \backslash A) \underset{\text{b)}}{\overset{\text{iii)}}{\in}} \mathcal{A}_\uplus$.

2.2) Extend μ_0 from $\mathcal{A}$ to its generated ring $\mathcal{A}_\uplus$ by $\mu_0(\biguplus_{i=1}^n A_i) := \sum_{i=1}^n \mu_0(A_i)$. This extension is unique as long as it is well-defined. To show well-definedness, let $\biguplus_{i=1}^n A_i = \biguplus_{j=1}^m B_j$, $A_i, B_j \in \mathcal{A}$. Then $A_i = A_i \cap (\biguplus_{j=1}^m B_j) \underset{\text{distr.}}{=} \biguplus_{j=1}^m (A_i \cap B_j)$. By additivity of μ_0 on $\mathcal{A}$, $\mu_0(A_i) = \sum_{j=1}^m \mu_0(A_i \cap B_j)$. Hence, $\sum_{i=1}^n \mu_0(A_i) = \sum_{i=1}^n \sum_{j=1}^m \mu_0(A_i \cap B_j) \underset{A_i, B_j}{\overset{\text{switch role of}}{=}} \dots = \sum_{j=1}^m \mu_0(B_j)$.

2.3) We now show that μ_0 is σ-additive on $\mathcal{A}_\uplus$, which implies that μ_0 is a premeasure on $\mathcal{A}_\uplus$.

- Let $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{A}_\uplus$, $A_i \cap A_j = \emptyset \; \forall\, i \neq j$, and $\biguplus_{i=1}^\infty A_i \in \mathcal{A}_\uplus \underset{\text{def. } \mathcal{A}_\uplus}{\Rightarrow} \forall\, i \in \mathbb{N}\; \exists\, n_i \in \mathbb{N} : (A_{i,j})_{j=1}^{n_i} \subseteq \mathcal{A}$ and $A_i = \biguplus_{j=1}^{n_i} A_{i,j}$, where $A_{i,j} \cap A_{k,l} = \emptyset$ $\forall\, i \neq k, \forall\, j, l$. By enumerating $(A_{i,j})$, $\exists$ pairwise disjoint $(\tilde{A}_k)_{k=1}^\infty \subseteq \mathcal{A}$ and a sequence of integers $0 =: k_0 < k_1 < k_2 \ldots$ such that $A_i = \biguplus_{j=k_{i-1}+1}^{k_i} \tilde{A}_j$, $i \in \mathbb{N}$. This implies that $\biguplus_{i=1}^\infty A_i = \biguplus_{i=1}^\infty \biguplus_{j=k_{i-1}+1}^{k_i} \tilde{A}_j$.
- Since $\biguplus_{i=1}^\infty A_i \in \mathcal{A}_\uplus$, there must exist $n \in \mathbb{N}$ and a partition $\{I_i\}_{i=1}^n$ of $\mathbb{N}$ such that $\biguplus_{i=1}^\infty A_i = \biguplus_{i=1}^n \biguplus_{j\in I_i} \tilde{A}_j$ for $\biguplus_{j\in I_i} \tilde{A}_j \in \mathcal{A}$.
- Therefore,

$$\mu_0\Big(\biguplus_{i=1}^\infty A_i\Big) = \mu_0\Big(\biguplus_{i=1}^n \biguplus_{j\in I_i} \tilde{A}_j\Big) \underset{\text{by 2.2)}}{\overset{\text{def. } \mu_0|_{\mathcal{A}_\uplus}}{=}} \sum_{i=1}^n \mu_0\Big(\biguplus_{j\in I_i} \tilde{A}_j\Big)$$

$$\underset{\text{by D. 2.33 iii)}}{\overset{\mu_0|_{\mathcal{A}}\ \sigma\text{-add.}}{=}} \sum_{i=1}^n \sum_{j\in I_i} \mu_0(\tilde{A}_j) \underset{\text{partitions } \mathbb{N}}{\overset{\{I_i\}_{i=1}^n}{=}} \sum_{i=1}^\infty \sum_{j=k_{i-1}+1}^{k_i} \mu_0(\tilde{A}_j) \underset{\text{by 2.2)}}{\overset{\text{def. } \mu_0|_{\mathcal{A}_\uplus}}{=}} \sum_{i=1}^\infty \mu_0(A_i).$$

2.4) One can show that the premeasure μ_0 on the ring $\mathcal{A}_\uplus$ is monotone and σ-subadditive; due to the ring properties, this works similarly as the proofs of P. 2.27 2) and 3). We can now show that $\mu^*|_{\mathcal{A}} = \mu_0|_{\mathcal{A}}$.

$\forall \{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{A} : \bigcup_{i=1}^\infty A_i \supseteq A \in \mathcal{A}$, we have $\mu_0(A) \underset{\text{distr.}}{=} \mu_0(\bigcup_{i=1}^\infty (A_i \cap A)) \underset{\sigma\text{-subadd.}}{\leq} \sum_{i=1}^\infty \mu_0(A_i \cap A) \underset{\text{def.}}{=} \lim_{n\to\infty} \sum_{i=1}^n \mu_0(A_i \cap A) \underset{\text{mon.}}{\leq} \lim_{n\to\infty} \sum_{i=1}^n \mu_0(A_i) \underset{\text{def.}}{=} \sum_{i=1}^\infty \mu_0(A_i)$. Hence also for the infimum over all such $\{A_i\}_{i\in\mathbb{N}}$, we have

$$\mu_0(A) \underset{\forall \{A_i\}_{i\in\mathbb{N}}}{\leq} \inf_{\substack{A \subseteq \bigcup_{i=1}^\infty A_i, \\ A_i \in \mathcal{A}}} \sum_{i=1}^\infty \mu_0(A_i) \underset{\text{def.}}{=} \mu^*(A).$$

And the cover $\{A, \emptyset, \emptyset, \ldots\} \subseteq \mathcal{A}$ implies that $\mu^*(A) \underset{\text{inf}}{\leq} \mu_0(A)$, so $\mu^*(A) = \mu_0(A)\ \forall A \in \mathcal{A}$.

3) We show that $\mathcal{A} \subseteq \mathcal{A}^*$, i.e. $\mu^*(B) = \mu^*(B \cap A) + \mu^*(B \cap A^c)\ \forall A \in \mathcal{A}, \forall B \subseteq \Omega$. Let $A \in \mathcal{A}$, $B \subseteq \Omega$. By definition of the infimum, $\forall \varepsilon > 0$, $\exists \{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{A} : \bigcup_{i=1}^\infty A_i \supseteq B$ and $\sum_{i=1}^\infty \mu_0(A_i) \leq \mu^*(B) + \varepsilon$. If $E_i := A_i \cap A \underset{\text{by D. 2.3 ii)}}{\overset{\mathcal{A}\ \pi\text{-sys.}}{\in}} \mathcal{A}$, $i \in \mathbb{N}$, then $A_i \backslash A = A_i \backslash E_i \underset{(*)}{\overset{\mathcal{A}\text{ semiring}}{=}} \biguplus_{k=1}^{n_i} C_{i,k}$ for some $n_i \in \mathbb{N}$ and disjoint $C_{i,1}, \ldots, C_{i,n_i} \in \mathcal{A}$. Hence $A_i = E_i \uplus A_i \backslash E_i = E_i \uplus \biguplus_{k=1}^{n_i} C_{i,k}$ and

$$B \cap A \underset{B \subseteq \bigcup_{i=1}^\infty A_i}{\subseteq} \left(\bigcup_{i=1}^\infty A_i\right) \cap A \underset{\text{distr.}}{=} \bigcup_{i=1}^\infty (A_i \cap A) \underset{\text{def.}}{=} \bigcup_{i=1}^\infty E_i,$$

$$B \cap A^c \underset{B \subseteq \bigcup_{i=1}^\infty A_i}{\subseteq} \left(\bigcup_{i=1}^\infty A_i\right) \cap A^c \underset{\text{distr.}}{=} \bigcup_{i=1}^\infty (\underbrace{A_i \cap A^c}_{= A_i \setminus A}) \underset{(*)}{=} \bigcup_{i=1}^\infty \biguplus_{k=1}^{n_i} C_{i,k}.$$

We thus obtain that

$$\mu^*(B \cap A) + \mu^*(B \cap A^c)$$

$$\underset{\text{by D. 2.34 ii)}}{\overset{\mu^* \text{ mon.}}{\leq}} \mu^*\left(\bigcup_{i=1}^\infty E_i\right) + \mu^*\left(\bigcup_{i=1}^\infty \biguplus_{k=1}^{n_i} C_{i,k}\right) \underset{\text{by D. 2.34 iii)}}{\overset{\mu^* \ \sigma\text{-subadd.}}{\leq}} \sum_{i=1}^\infty \mu^*(E_i) + \sum_{i=1}^\infty \sum_{k=1}^{n_i} \mu^*(C_{i,k})$$

$$\underset{\mu^*|_{\mathcal{A}} \underset{2)}{=} \mu_0|_{\mathcal{A}}}{=} \sum_{i=1}^\infty \mu_0(E_i) + \sum_{i=1}^\infty \sum_{k=1}^{n_i} \mu_0(C_{i,k}) = \sum_{i=1}^\infty \left(\mu_0(E_i) + \sum_{k=1}^{n_i} \mu_0(C_{i,k})\right)$$

$$\underset{\text{by D. 2.33 iii)}}{\overset{\mu_0 \text{ add.}}{=}} \sum_{i=1}^\infty \mu_0\left(E_i \uplus \biguplus_{k=1}^{n_i} C_{i,k}\right) = \sum_{i=1}^\infty \mu_0(A_i) \leq \mu^*(B) + \varepsilon.$$

Letting $\varepsilon \to 0+$, we obtain $\mu^*(B \cap A) + \mu^*(B \cap A^c) \leq \mu^*(B)$. Trivially,

$$\mu^*(B) = \mu^*((B \cap A) \uplus (B \cap A^c)) \underset{\text{by D. 2.34 iii)}}{\overset{\mu^* \text{ subadd.}}{\leq}} \mu^*(B \cap A) + \mu^*(B \cap A^c),$$

so we have equality and thus obtain that $A \underset{\text{def.}}{\in} \mathcal{A}^*$, so $\mathcal{A} \subseteq \mathcal{A}^*$.

4) We show that $\mathcal{A}^*$ is a σ-algebra (by showing it is a Dynkin system and a π-system; see P. 2.18 3)) and that μ^* is a measure on $\mathcal{A}^*$.

4.1) π-**system:** Let $A_1, A_2 \in \mathcal{A}^*$ and $B \subseteq \Omega$. Then $\underbrace{(A_1 \cap A_2)^c}_{\text{contained in at most 1}} \underset{\text{Venn}}{=} \underbrace{(A_1^c \cap A_2)}_{\text{strictly in } A_2} \uplus \underbrace{(A_1 \cap A_2^c)}_{\text{strictly in } A_1} \uplus \underbrace{(A_1^c \cap A_2^c)}_{\text{contained in neither}}$ implies that

$$\begin{aligned}
\mu^*(B) &\underset{\text{by D. 2.34 iii)}}{\overset{\mu^* \text{ subadd.}}{\leq}} \mu^*(B \cap (A_1 \cap A_2)) + \mu^*(B \cap (A_1 \cap A_2)^c) \\
&= \mu^*(B \cap A_1 \cap A_2) \\
&\quad + \mu^*((B \cap A_1^c \cap A_2) \uplus (B \cap A_1 \cap A_2^c) \uplus (B \cap A_1^c \cap A_2^c)) \\
&\underset{\text{by D. 2.34 iii)}}{\overset{\mu^* \text{ subadd.}}{\leq}} \mu^*(B \cap A_1 \cap A_2) + \mu^*(B \cap A_1^c \cap A_2) + \mu^*(B \cap A_1 \cap A_2^c) \\
&\quad + \mu^*(B \cap A_1^c \cap A_2^c) \\
&\underset{\text{reorder}}{=} \mu^*(B \cap A_2 \cap A_1) + \mu^*(B \cap A_2 \cap A_1^c) \\
&\quad + \mu^*(B \cap A_2^c \cap A_1) + \mu^*(B \cap A_2^c \cap A_1^c) \\
&\underset{B \cap A_2,\, B \cap A_2^c \subseteq \Omega}{\overset{A_1 \in \mathcal{A}^*}{=}} \mu^*(B \cap A_2) + \mu^*(B \cap A_2^c) \underset{B \subseteq \Omega}{\overset{A_2 \in \mathcal{A}^*}{=}} \mu^*(B),
\end{aligned}$$

so $A_1 \cap A_2$ satisfies $\mu^*(B) = \mu^*(B \cap (A_1 \cap A_2)) + \mu^*(B \cap (A_1 \cap A_2)^c)$ $\forall\, B \subseteq \Omega$ and thus $A_1 \cap A_2 \in \mathcal{A}^*$.

4.2) **Dynkin system:**

i) $\forall\, B \subseteq \Omega$, we have $\mu^*(B) \underset{\text{by def.}}{\overset{\mu^*(\emptyset) = 0}{=}} \mu^*(B \cap \Omega) + \mu^*(B \cap \emptyset)$, so $\Omega \in \mathcal{A}^*$.

ii) By definition of $\mathcal{A}^*$, $A \in \mathcal{A}^*$ iff $A^c \in \mathcal{A}^*$, so $A \in \mathcal{A}^* \Rightarrow A^c \in \mathcal{A}^*$.

iii) Let $\{A_i\}_{i\in\mathbb{N}} \subseteq \mathcal{A}^*$, $A_i \cap A_j = \emptyset \; \forall\, i \neq j$. With $\tilde{A}_n := \biguplus_{i=1}^n A_i \underset{\text{De Morgan}}{=} (\bigcap_{i=1}^n A_i^c)^c \underset{4.1)}{\overset{\text{ii)}}{\in}} \mathcal{A}^*$, $n \in \mathbb{N}$, and $\tilde{A}_\infty := \biguplus_{i=1}^\infty A_i$, we have $\forall\, B \subseteq \Omega$,

$$\mu^*(B \cap \tilde{A}_{n+1}) \underset{\tilde{A}_n \in \mathcal{A}^*}{=} \mu^*((B \cap \tilde{A}_{n+1}) \cap \tilde{A}_n) + \mu^*((B \cap \tilde{A}_{n+1}) \cap \tilde{A}_n^c)$$

$$\underset{\substack{\tilde{A}_{n+1} \cap \tilde{A}_n = \tilde{A}_n \\ \tilde{A}_{n+1} \cap \tilde{A}_n^c = A_{n+1}}}{=} \mu^*(B \cap \tilde{A}_n) + \mu^*(B \cap A_{n+1}), \quad n \in \mathbb{N}.$$

This implies, inductively, that $\mu^*(B \cap \tilde{A}_n) \underset{(*)}{=} \sum_{i=1}^n \mu^*(B \cap A_i)$, so

$$\mu^*(B \cap \tilde{A}_\infty) + \mu^*(B \cap \tilde{A}_\infty^c) \underset{\text{by D. 2.34 iii)}}{\overset{\mu^* \text{ subadd.}}{\geq}} \mu^*((B \cap \tilde{A}_\infty) \uplus (B \cap \tilde{A}_\infty^c))$$

$$= \mu^*(B) \underset{\tilde{A}_n \in \mathcal{A}^*}{=} \mu^*(B \cap \tilde{A}_n) + \mu^*(B \cap \tilde{A}_n^c)$$

$$\underset{\mu^* \text{ mon. by def.}}{\overset{\tilde{A}_n \subseteq \tilde{A}_\infty}{\geq}} \mu^*(B \cap \tilde{A}_n) + \mu^*(B \cap \tilde{A}_\infty^c) \underset{(*)}{=} \sum_{i=1}^n \mu^*(B \cap A_i) + \mu^*(B \cap \tilde{A}_\infty^c).$$

As this inequality holds $\forall\, n \in \mathbb{N}$, take the limit $n \to \infty$ to see that

$$\mu^*(B \cap \tilde{A}_\infty) + \mu^*(B \cap \tilde{A}_\infty^c) \geq \mu^*(B) \geq \sum_{i=1}^\infty \mu^*(B \cap A_i) + \mu^*(B \cap \tilde{A}_\infty^c)$$

$$\underset{\text{by D. 2.34 iii)}}{\overset{\mu^* \ \sigma\text{-subadd.}}{\geq}} \mu^*\left(B \cap \biguplus_{i=1}^\infty A_i\right) + \mu^*(B \cap \tilde{A}_\infty^c) \underset{\text{def. } \tilde{A}_\infty}{=} \mu^*(B \cap \tilde{A}_\infty) + \mu^*(B \cap \tilde{A}_\infty^c)$$

and thus equality, so $\tilde{A}_\infty \in \mathcal{A}^*$ and thus $\mathcal{A}^*$ is a Dynkin system. By 4.1), $\mathcal{A}^*$ is a σ-algebra. For $B = \tilde{A}_\infty$, we get σ-additivity of μ^*:

$$\mu^*\left(\biguplus_{i=1}^{\infty} A_i\right) = \mu^*(\tilde{A}_\infty) \underset{(*)}{=} \sum_{i=1}^{\infty} \mu^*(\tilde{A}_\infty \cap A_i) + \mu^*(\tilde{A}_\infty \cap \tilde{A}_\infty^c)$$
$$= \sum_{i=1}^{\infty} \mu^*\left(\left(\biguplus_{i=1}^{\infty} A_i\right) \cap A_i\right) + 0 = \sum_{i=1}^{\infty} \mu^*(A_i).$$

4.3) μ^* is a measure on the σ-algebra $\mathcal{A}^*$:

i) $\mu^* : \mathcal{P}(\Omega) \to [0, \infty] \Rightarrow \mu^* : \mathcal{A}^* \to [0, \infty]$;

ii) $\mu^*(\emptyset) \underset{\text{D. 2.34 i)}}{=} 0$; and

iii) μ^* is σ-additive on $\mathcal{A}^*$, see the last statement in 4) iii).

5) $\mathcal{A} \underset{3)}{\subseteq} \mathcal{A}^*$. By 4), $\mathcal{A}^*$ is a σ-algebra $\underset{\sigma \text{ smallest}}{\Rightarrow} \sigma(\mathcal{A}) \subseteq \mathcal{A}^* \underset{4)}{\Rightarrow} \mu := \mu^*|_{\sigma(\mathcal{A})}$ is a measure. And $\mu|_{\mathcal{A}} \underset{\text{def.}}{=} \mu^*|_{\mathcal{A}} \underset{2)}{=} \mu_0|_{\mathcal{A}}$, so μ extends μ_0.

6) By ass., μ_0 is σ-finite on $\mathcal{A} \underset{\text{def.}}{\Rightarrow} \mu^*$ and thus μ are σ-finite. Furthermore, as a semiring, $\mathcal{A}$ is a π-system $\underset{\text{P. 2.28}}{\Rightarrow}$ Another σ-finite measure ν on $\sigma(\mathcal{A})$ with $\nu|_{\mathcal{A}} = \mu|_{\mathcal{A}}$ must coincide with μ on $\sigma(\mathcal{A})$, so μ is unique on $\sigma(\mathcal{A})$. $\square$

By T. 2.32 we can always assume the extension μ to be complete, which we do.

2.6 Borel measures on $\mathbb{R}^d$

Question: How can we use Carathéodory's extension theorem to construct Borel measures on $\mathbb{R}^d$?

- At the core of the construction lie functions $F : \mathbb{R}^d \to \mathbb{R}$ that are right-continuous and increasing in a specific way.
- $F : \mathbb{R}^d \to \mathbb{R}$ is *right-continuous* if $F(\boldsymbol{x}) = \lim_{\boldsymbol{h}\to\boldsymbol{0}+} F(\boldsymbol{x}+\boldsymbol{h}) =: F(\boldsymbol{x}+)$ $\forall\, \boldsymbol{x} \in \mathbb{R}^d$.
- One also frequently utilizes F that are *grounded*, i.e. $\forall\, j \in \{1,\dots,d\}$ one has $\lim_{x_j\to-\infty} F(\boldsymbol{x}) = 0$. This will be crucial for constructing Borel probability measures on $\mathbb{R}^d$.

Definition 2.36 (d-increasing)
$F : \mathbb{R}^d \to \mathbb{R}$ is *d-increasing* if $\Delta_{(\boldsymbol{a},\boldsymbol{b}]} F \geq 0\ \forall\, \boldsymbol{a} \leq \boldsymbol{b}$, where the *$F$-volume* is

$$\Delta_{(\boldsymbol{a},\boldsymbol{b}]} F := \sum_{\boldsymbol{i}\in\{0,1\}^d} (-1)^{\sum_{j=1}^d i_j} F(\underbrace{a_1^{i_1} b_1^{1-i_1}}_{=\begin{cases} b_1, & i_1 = 0, \\ a_1, & i_1 = 1. \end{cases}}, \dots, a_d^{i_d} b_d^{1-i_d}).$$

- For $d = 1$: $\Delta_{(a_1,b_1]}F = F(b_1) - F(a_1)$, i.e. $\Delta_{(a_1,b_1]}F \geq 0$ implies that F is *increasing* on $\mathbb{R}$ ($F \nearrow$) in the usual sense.
- For $d = 2$: $\Delta_{(\boldsymbol{a},\boldsymbol{b}]}F = \underbrace{F(b_1, b_2) - F(a_1, b_2)}_{=\Delta_{(a_1,b_1]}F(x_1,b_2)} - (\underbrace{F(b_1, a_2) - F(a_1, a_2)}_{=\Delta_{(a_1,b_1]}F(x_1,a_2)})$ $= \Delta_{(a_2,b_2]}\Delta_{(a_1,b_1]}F$.

Question: How can we better understand F-volumes?

For $J \subseteq \{1, \dots, d\}$, let $\boldsymbol{x}_J = (x_j)_{j \in J}$ and $\boldsymbol{x}_{J^c} = (x_j)_{j \notin J}$. For $x \in \mathbb{R}$, let ${}_d x := (x, \dots, x) \in \mathbb{R}^d$. And for $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d$, let

$$\boldsymbol{x}_{J \leftarrow \boldsymbol{y}_J} = \begin{cases} x_j, & j \notin J, \\ y_j, & j \in J; \end{cases}$$

for $J = \{j\}$, we simply write $\boldsymbol{x}_{j \leftarrow y_j} = (x_1, \dots, x_{j-1}, y_j, x_{j+1}, \dots, x_d)$.

Lemma 2.37 (Understanding F-volumes)
Let $F : \mathbb{R}^d \to \mathbb{R}$ be d-increasing.

1) Then $\Delta_{(\boldsymbol{a},\boldsymbol{b}]}F = \Delta_{(a_d,b_d]} \dots \Delta_{(a_1,b_1]}F$, or any permutation of first-order differences.
2) F-volumes are monotone, i.e. $(\boldsymbol{a}, \boldsymbol{b}] \subseteq (\boldsymbol{c}, \boldsymbol{d}]$ implies $\Delta_{(\boldsymbol{a},\boldsymbol{b}]}F \leq \Delta_{(\boldsymbol{c},\boldsymbol{d}]}F$.

3) If $F(\boldsymbol{x}) = \prod_{j=1}^d F_j(x_j)$, then $\Delta_{(\boldsymbol{a},\boldsymbol{b}]}F = \prod_{j=1}^d (F_j(b_j) - F_j(a_j))$.
4) If F is grounded, then $\Delta_{(-\infty,\boldsymbol{x}]}F = F(\boldsymbol{x})$, $\boldsymbol{x} \in \mathbb{R}^d$, where $\Delta_{(-\infty,\boldsymbol{x}]}F := \lim_{\boldsymbol{a}\to-\infty} \Delta_{(\boldsymbol{a},\boldsymbol{x}]}F$.
5) If F is grounded and $\emptyset \neq J \subsetneq \{1,\dots,d\}$, then $\Delta_{(\boldsymbol{a}_J,\boldsymbol{b}_J]}F(\boldsymbol{x}) \geq 0 \;\forall\, \boldsymbol{x}_{J^c} \in \mathbb{R}^{|J^c|}$. In particular, for $J = \{j\}$, $j = 1,\dots,d$, F is componentwise increasing.

Proof.

1) Proof by induction:

$d = 1$: $\Delta_{(a,b]}F \underset{\text{def.}}{=} F(b) - F(a) = \Delta_{(a,b]}F$ ✓

$d - 1 \Rightarrow d$:

$$\begin{aligned}
\Delta_{(\boldsymbol{a},\boldsymbol{b}]}F &\underset{\text{def.}}{=} \sum_{\boldsymbol{i}\in\{0,1\}^d} (-1)^{\sum_{j=1}^d i_j} F(a_1^{i_1} b_1^{1-i_1}, \dots, a_d^{i_d} b_d^{1-i_d}) \\
&= \sum_{\boldsymbol{i}\in\{0,1\}^d:\, i_d=0} (-1)^{\sum_{j=1}^{d-1} i_j} F(a_1^{i_1} b_1^{1-i_1}, \dots, a_{d-1}^{i_{d-1}} b_{d-1}^{1-i_{d-1}}, b_d) \\
&\quad - \sum_{\boldsymbol{i}\in\{0,1\}^d:\, i_d=1} (-1)^{\sum_{j=1}^{d-1} i_j} F(a_1^{i_1} b_1^{1-i_1}, \dots, a_{d-1}^{i_{d-1}} b_{d-1}^{1-i_{d-1}}, a_d) \\
&\underset{\text{i.h.}}{=} \Delta_{(a_{d-1},b_{d-1}]} \dots \Delta_{(a_1,b_1]} F(x_1 \dots, x_{d-1}, b_d)
\end{aligned}$$

$$
\begin{aligned}
&- \Delta_{(a_{d-1},b_{d-1}]} \dots \Delta_{(a_1,b_1]} F(x_1 \dots, x_{d-1}, a_d) \\
&= \Delta_{(a_d,b_d]} \Delta_{(a_{d-1},b_{d-1}]} \dots \Delta_{(a_1,b_1]} F.
\end{aligned}
$$

This holds for any permutation of first-order differences since addition is commutative.

2) For all $j \in \{1, \dots, d\}$, we have

$$
\Delta_{(a_j,b_j]} \Delta_{(\boldsymbol{a}_{j^c},\boldsymbol{b}_{j^c}]} F \underset{1)}{=} \Delta_{(\boldsymbol{a},\boldsymbol{b}]} F \underset{d\text{-incr.}}{\geq} 0, \quad a_j \leq b_j
$$

Therefore, $x_j \mapsto \Delta_{(\boldsymbol{a}_{j^c},\boldsymbol{b}_{j^c}]} F(\boldsymbol{x})$ is $\nearrow$ $(*)$. Since $(\boldsymbol{a}, \boldsymbol{b}] \subseteq (\boldsymbol{c}, \boldsymbol{d}]$, we thus have that

$$
\begin{aligned}
\Delta_{(\boldsymbol{a},\boldsymbol{b}]} F &\underset{1)}{=} \Delta_{(a_d,b_d]} \Delta_{(a_{d-1},b_{d-1}]} \dots \Delta_{(a_1,b_1]} F \underset{(*)}{\leq} \Delta_{(c_d,d_d]} \Delta_{(a_{d-1},b_{d-1}]} \dots \Delta_{(a_1,b_1]} F \\
&\underset{1)}{=} \Delta_{(a_{d-1},b_{d-1}]} \Delta_{(c_d,d_d]} \dots \Delta_{(a_1,b_1]} F \overset{(*)}{\underset{1)}{\leq}} \dots \overset{(*)}{\underset{1)}{\leq}} \Delta_{(\boldsymbol{c},\boldsymbol{d}]} F.
\end{aligned}
$$

3) We have

$$
\begin{aligned}
\Delta_{(\boldsymbol{a},\boldsymbol{b}]} F &\underset{1)}{=} \Delta_{(a_d,b_d]} \dots \Delta_{(a_2,b_2]} \Delta_{(a_1,b_1]} F_1(x_1) F_2(x_2) \cdot \dots \cdot F_d(x_d) \\
&= \Delta_{(a_d,b_d]} \dots \Delta_{(a_2,b_2]} (F_1(b_1) - F_1(a_1)) \prod_{j=2}^{d} F_j(x_j)
\end{aligned}
$$

$$= (F_1(b_1) - F_1(a_1))\Delta_{(a_d,b_d]} \dots \Delta_{(a_2,b_2]} \prod_{j=2}^{d} F_j(x_j)$$

$$\underset{d-1 \text{ times}}{\overset{\text{apply}}{=}} \prod_{j=1}^{d} (F_j(b_j) - F_j(a_j)). \qquad \square$$

4) For $\boldsymbol{x} \in \mathbb{R}^d$, we have

$$\Delta_{(-\infty,\boldsymbol{x}]}F = \lim_{\boldsymbol{a}\to-\infty} \Delta_{(\boldsymbol{a},\boldsymbol{x}]}F \underset{\text{def.}}{=} \lim_{\boldsymbol{a}\to-\infty} \sum_{\boldsymbol{i}\in\{0,1\}^d} (-1)^{\sum_{j=1}^d i_j} F(a_1^{i_1}x_1^{1-i_1}, \dots, a_d^{i_d}x_d^{1-i_d})$$

$$= \sum_{\boldsymbol{i}\in\{0,1\}^d} (-1)^{\sum_{j=1}^d i_j} \underbrace{\lim_{\boldsymbol{a}\to-\infty} F(a_1^{i_1}x_1^{1-i_1}, \dots, a_d^{i_d}x_d^{1-i_d})}_{= \begin{cases} 0, & \exists\, j : i_j = 1 \text{ by groundedness}, \\ F(\boldsymbol{x}), & \boldsymbol{i} = \boldsymbol{0}, \end{cases}} = F(\boldsymbol{x}).$$

5) Expanding $\boldsymbol{a}_J$ to $\boldsymbol{a}$ such that $\boldsymbol{a}_{J^c} = -\boldsymbol{\infty}$, and $\boldsymbol{b}_J$ to $\boldsymbol{b}$ such that $\boldsymbol{b}_{J^c} = \boldsymbol{x}_{J^c}$, we have $\Delta_{(\boldsymbol{a}_J,\boldsymbol{b}_J]}F(\boldsymbol{x}) \underset{4)}{=} \Delta_{(\boldsymbol{a},\boldsymbol{b}]}F \underset{d\text{-incr.}}{\geq} 0 \;\forall\, \boldsymbol{x}_{J^c} \in \mathbb{R}^{|J^c|}$.

Before we can construct Borel measures on $\mathbb{R}^d$, we need one more auxiliary result.

Lemma 2.38 (Representation of differences involving multiple sets)
Let $\mathcal{A}$ be a semiring and $A, A_1, \dots, A_n \in \mathcal{A}$. Then $\exists\, m \in \mathbb{N}$ and pairwise disjoint $B_1, \dots, B_m \in \mathcal{A}$ such that $A \backslash \bigcup_{i=1}^n A_i = \biguplus_{j=1}^m B_j$.

Proof. Proof by induction:

$n = 1 : A \backslash A_1 \underset{\text{D. 2.3 iii)}}{=} \biguplus_{j=1}^m B_j$ ✓

$n - 1 \Rightarrow n$: We have

$$A \backslash \bigcup_{i=1}^n A_i = \left(A \backslash \bigcup_{i=1}^{n-1} A_i \right) \backslash A_n \underset{\text{hypo.}}{=} \left(\biguplus_{k=1}^p C_k \right) \backslash A_n \underset{\text{distr.}}{=} \biguplus_{k=1}^p (C_k \backslash A_n) \underset{\text{D. 2.3 iii)}}{=} \biguplus_{k=1}^p \biguplus_{l=1}^{p_k} C_{k,l}$$

for pairwise disjoint $\{C_k\}_{k=1}^p, \{C_{k,l}\}_{k,l=1}^{p,p_k} \subseteq \mathcal{A}$, which is of the required form. □

Lemma
Let μ_0 be an additive premeasure on a semiring $\mathcal{A}$.

1) If $A_1, \dots, A_n \in \mathcal{A}$ are pairwise disjoint and $A \in \mathcal{A}$ such that $\biguplus_{i=1}^n A_i \subseteq A$, then $\sum_{i=1}^n \mu_0(A_i) \leq \mu_0(A)$.
2) If $A_1, \dots, A_n, A \in \mathcal{A}$ such that $A \subseteq \bigcup_{i=1}^n A_i$, then $\mu_0(A) \leq \sum_{i=1}^n \mu_0(A_i)$.

Proof.

1) By L. 2.38, $A \setminus \biguplus_{i=1}^n A_i = \biguplus_{j=1}^m B_j$ for pairwise disjoint $B_1, \dots, B_m \in \mathcal{A}$ and $A_1, \dots, A_n, B_1, \dots, B_m$ are pairwise disjoint. Therefore, $\mu_0(A) \underset{\text{add.}}{=} \sum_{i=1}^n \mu_0(A_i) + \sum_{j=1}^m \mu_0(B_j) \underset{\mu_0 \geq 0}{\geq} \sum_{i=1}^n \mu_0(A_i)$.

2) For $i = 1, \dots, n$, let $A_i' = A \cap A_i \in \mathcal{A}$ and $A_i'' = A_i' \setminus \bigcup_{k=1}^{i-1} A_k'$. Then $A \underset{A \subseteq \bigcup_{i=1}^n A_i}{=} A \cap \bigcup_{i=1}^n A_i \underset{\text{distr.}}{=} \bigcup_{i=1}^n A_i' = \biguplus_{i=1}^n A_i''$. For all $i = 1, \dots, n$, $A_i'' \underset{\text{L. 2.38}}{=} \biguplus_{k=1}^{n_i} B_{i,k}$ for pairwise disjoint $B_{i,k} \in \mathcal{A}$, $k = 1, \dots, n_i$, and thus $A = \biguplus_{n=1}^n \biguplus_{k=1}^{n_i} B_{i,k}$. Since $A_1'', \dots, A_n''$ are pairwise disjoint, so are $B_{i,k}$ $\forall\, i, k$. Also, $\biguplus_{k=1}^{n_i} B_{i,k} = A_i'' \subseteq A_i' \subseteq A_i$ $\forall\, i$, so $\sum_{k=1}^{n_i} \mu_0(B_{i,k}) \underset{\text{add.}}{=} \mu_0(\biguplus_{k=1}^{n_i} B_{i,k}) \underset{1)}{\leq} \mu_0(A_i)$. Therefore, $\mu_0(A) \underset{\text{add.}}{=} \sum_{i=1}^n \sum_{k=1}^{n_i} \mu_0(B_{i,k}) \leq \sum_{i=1}^n \mu_0(A_i)$. $\square$

Theorem 2.39 (Construction of Borel measures on $\mathbb{R}^d$)
If $F : \mathbb{R}^d \to \mathbb{R}$ is d-increasing and right-continuous, $\exists!$ Borel measure λ_F such that $\lambda_F((\boldsymbol{a}, \boldsymbol{b}]) = \Delta_{(\boldsymbol{a},\boldsymbol{b}]} F$, $\boldsymbol{a} \leq \boldsymbol{b}$.

Proof.

- $\mathcal{A} := \{(\boldsymbol{a}, \boldsymbol{b}] : -\infty < \boldsymbol{a} \leq \boldsymbol{b} < \infty\}$ is a semiring on $\mathbb{R}^d$:
  i) For any $\boldsymbol{a} \in \mathbb{R}^d$, $\emptyset = (\boldsymbol{a}, \boldsymbol{a}] \in \mathcal{A}$.

ii) For $i = 1, 2$, $-\infty < \boldsymbol{a}_i \leq \boldsymbol{b}_i < \infty$, we have

$$(\boldsymbol{a}_1, \boldsymbol{b}_1] \cap (\boldsymbol{a}_2, \boldsymbol{b}_2] = \left(\begin{pmatrix} \max_{i=1,2}\{a_{i,1}\} \\ \vdots \\ \max_{i=1,2}\{a_{i,d}\} \end{pmatrix}, \begin{pmatrix} \min_{i=1,2}\{b_{i,1}\} \\ \vdots \\ \min_{i=1,2}\{b_{i,d}\} \end{pmatrix} \right] \in \mathcal{A},$$

interpreted as $\emptyset \in \mathcal{A}$ if $\exists\, j \in \{1, \ldots, d\} : \max_{i=1,2}\{a_{i,j}\} \geq \min_{i=1,2}\{b_{i,j}\}$.

iii) The difference of two hypercubes $(\boldsymbol{a}_1, \boldsymbol{b}_1]$, $(\boldsymbol{a}_2, \boldsymbol{b}_2]$ is the union of (at most $2d$, so) finitely many hypercubes and thus in $\mathcal{A}$.

- We now show that $\mu_0((\boldsymbol{a}, \boldsymbol{b}]) := \Delta_{(\boldsymbol{a},\boldsymbol{b}]} F$ is a σ-finite premeasure on $\mathcal{A}$:

i) F d-increasing $\Rightarrow \mu_0 : \mathcal{A} \to [0, \infty]$.

ii) $\mu_0(\emptyset) = \mu_0((\boldsymbol{a}, \boldsymbol{a}]) \underset{\text{def.}}{=} \Delta_{(\boldsymbol{a},\boldsymbol{a}]} F \underset{\text{def.}}{=} \sum_{\boldsymbol{i} \in \{0,1\}^d} (-1)^{\sum_{j=1}^d i_j} F(a_1^{i_1} a_1^{1-i_1}, \ldots, a_d^{i_d} a_d^{1-i_d}) = F(\boldsymbol{a}) \sum_{\boldsymbol{i} \in \{0,1\}^d} (-1)^{\sum_{j=1}^d i_j} = 0$.

iii) σ-additivity of μ_0: If $\mathcal{A} \ni (\boldsymbol{a}, \boldsymbol{b}] = \biguplus_{i=1}^n (\boldsymbol{a}_i, \boldsymbol{b}_i]$, then $\mu_0(\biguplus_{i=1}^n (\boldsymbol{a}_i, \boldsymbol{b}_i]) \underset{\text{def.}}{=} \mu_0((\boldsymbol{a}, \boldsymbol{b}]) \underset{\text{def.}}{=} \Delta_{(\boldsymbol{a},\boldsymbol{b}]} F \overset{\text{mult.}}{\underset{\text{tele.}}{=}} \sum_{i=1}^n \Delta_{(\boldsymbol{a}_i,\boldsymbol{b}_i]} F \underset{\text{def.}}{=} \sum_{i=1}^n \mu_0((\boldsymbol{a}_i, \boldsymbol{b}_i])$, so μ_0 is additive. Let $\biguplus_{i=1}^\infty (\boldsymbol{a}_i, \boldsymbol{b}_i] = (\boldsymbol{a}, \boldsymbol{b}] \in \mathcal{A}$.

  - $\sum_{i=1}^\infty \mu_0((\boldsymbol{a}_i, \boldsymbol{b}_i]) \leq \mu_0((\boldsymbol{a}, \boldsymbol{b}])$: $\forall\, n \in \mathbb{N}$, $\biguplus_{i=1}^n (\boldsymbol{a}_i, \boldsymbol{b}_i] \subseteq \biguplus_{i=1}^\infty (\boldsymbol{a}_i, \boldsymbol{b}_i] = (\boldsymbol{a}, \boldsymbol{b}] \underset{\text{L. 1)}}{\Rightarrow} \sum_{i=1}^n \mu_0((\boldsymbol{a}_i, \boldsymbol{b}_i]) \leq \mu_0((\boldsymbol{a}, \boldsymbol{b}]) \underset{n \to \infty}{\Rightarrow}$ ✓.

- $\mu_0((\boldsymbol{a},\boldsymbol{b}]) \leq \sum_{i=1}^\infty \mu_0((\boldsymbol{a}_i,\boldsymbol{b}_i])$: F right-continuous $\Rightarrow \boldsymbol{x} \mapsto \Delta_{(\boldsymbol{x},\boldsymbol{b}]}F = \mu_0((\boldsymbol{x},\boldsymbol{b}])$ is right-continuous $\Rightarrow$ For $\varepsilon > 0$, $\exists\, \tilde{\boldsymbol{a}} \in (\boldsymbol{a},\boldsymbol{b}] : \mu_0((\boldsymbol{a},\boldsymbol{b}]) \underset{(*)}{\leq} \mu_0((\tilde{\boldsymbol{a}},\boldsymbol{b}]) + \varepsilon$. Similarly, $\boldsymbol{x} \mapsto \mu_0((\boldsymbol{a},\boldsymbol{x}])$ is right-continuous, so $\forall i \in \mathbb{N}\ \exists\, \tilde{\boldsymbol{b}}_i \in (\boldsymbol{b}_i,\boldsymbol{b}] : \mu_0((\boldsymbol{a}_i,\tilde{\boldsymbol{b}}_i]) \underset{(**)}{\leq} \mu_0((\boldsymbol{a}_i,\boldsymbol{b}_i]) + \frac{\varepsilon}{2^i}$. We also have $[\tilde{\boldsymbol{a}},\boldsymbol{b}] \subseteq (\boldsymbol{a},\boldsymbol{b}] \underset{\text{ass.}}{=} \biguplus_{i=1}^\infty (\boldsymbol{a}_i,\boldsymbol{b}_i] \subseteq \bigcup_{i=1}^\infty (\boldsymbol{a}_i,\tilde{\boldsymbol{b}}_i) \underset{\text{Heine–Borel}}{\Rightarrow} \exists\, n \in \mathbb{N} : [\tilde{\boldsymbol{a}},\boldsymbol{b}] \subseteq \bigcup_{k=1}^n (\boldsymbol{a}_{i_k},\tilde{\boldsymbol{b}}_{i_k})$. This implies that $(\tilde{\boldsymbol{a}},\boldsymbol{b}] \subseteq [\tilde{\boldsymbol{a}},\boldsymbol{b}] \subseteq \bigcup_{k=1}^n (\boldsymbol{a}_{i_k},\tilde{\boldsymbol{b}}_{i_k}) \subseteq \bigcup_{k=1}^n (\boldsymbol{a}_{i_k},\tilde{\boldsymbol{b}}_{i_k}]$, so $\mu_0((\tilde{\boldsymbol{a}},\boldsymbol{b}]) \underset{\text{L. 2)}}{\leq} \sum_{k=1}^n \mu_0((\boldsymbol{a}_{i_k},\tilde{\boldsymbol{b}}_{i_k}])$. Hence

$$\begin{aligned}\mu_0((\boldsymbol{a},\boldsymbol{b}]) &\underset{(*)}{\leq} \mu_0((\tilde{\boldsymbol{a}},\boldsymbol{b}]) + \varepsilon \leq \sum_{k=1}^n \mu_0((\boldsymbol{a}_{i_k},\tilde{\boldsymbol{b}}_{i_k}]) + \varepsilon \\ &\underset{(**)}{\leq} \sum_{k=1}^n \left(\mu_0((\boldsymbol{a}_{i_k},\boldsymbol{b}_{i_k}]) + \frac{\varepsilon}{2^{i_k}}\right) + \varepsilon \underset{\mu_0 \geq 0}{\leq} \sum_{i=1}^\infty \mu_0((\boldsymbol{a}_i,\boldsymbol{b}_i]) + 2\varepsilon.\end{aligned}$$

$\underset{\varepsilon \to 0+}{\Rightarrow}$ ✓$\Rightarrow \mu_0$ is σ-additive $\Rightarrow \mu_0$ is a premeasure on $\mathcal{A}$.

- With $A_i = ({}_d i - \mathbf{1}, {}_d i] \in \mathcal{A}$, $i \in \mathbb{N}$, we have $\mathbb{R}^d = \bigcup_{i\in\mathbb{N}} A_i$ and $\mu_0(A_i) = \Delta_{({}_d i - \mathbf{1}, {}_d i]}F < \infty\ \forall\, i$, so μ_0 is σ-finite.
- By Carathéodory's extension theorem, $\exists!$ measure λ_F on $\sigma(\mathcal{A}) \underset{\text{R. 2.24 1)}}{=} \mathcal{B}(\mathbb{R}^d)$: $\lambda_F\big|_{\mathcal{A}} = \mu_0$, i.e. $\lambda_F((\boldsymbol{a},\boldsymbol{b}]) = \mu_0((\boldsymbol{a},\boldsymbol{b}]) = \Delta_{(\boldsymbol{a},\boldsymbol{b}]}F$, $-\infty < \boldsymbol{a} \leq \boldsymbol{b} < \infty$. $\square$

- As mentioned before, we can assume λ_F to be complete. It is known as *Lebesgue–Stieltjes measure* associated to F.
- If $F(\boldsymbol{x}) = \prod_{j=1}^d x_j$, $\boldsymbol{x} \in \mathbb{R}^d$, then $\lambda := \lambda_F$ is the *Lebesgue measure* on $\mathbb{R}^d$. Its domain is the completion $\bar{\mathcal{B}}(\mathbb{R}^d)$, known as *Lebesgue σ-algebra*. Sets in $\bar{\mathcal{B}}(\mathbb{R}^d)$ are *Lebesgue measurable* (or *Lebesgue sets*). Clearly, $\mathcal{B}(\mathbb{R}^d) \subseteq \bar{\mathcal{B}}(\mathbb{R}^d)$. One can also show that $\mathcal{B}(\mathbb{R}^d) \neq \bar{\mathcal{B}}(\mathbb{R}^d)$; see E. 3.9 later.
- The Lebesgue measure satisfies $\lambda((\boldsymbol{a}, \boldsymbol{b}]) \underset{\text{L. 2.37 3)}}{=} \prod_{j=1}^d (b_j - a_j)$. It thus assigns hypercubes their volume as we set out to construct; one can also show its invariance wrt translations, rotations and reflections. For general F, λ_F can be interpreted as assigning hypercubes their volume distorted by F.
- Since $\mathbb{R}^d = \prod_{j=1}^d \mathbb{R}$ is a product space, we can also equip $\mathbb{R}^d$ with the product-σ-algebra $\bigotimes_{j=1}^d \mathcal{F}_j \underset{\text{P. 2.16}}{=} \sigma(\prod_{j=1}^d B_j : B_j \in \mathcal{B}(\mathbb{R}))$, which, by R. 2.24 1), is $\mathcal{B}(\mathbb{R}^d)$. On $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, we can then consider the product measure $\mu = \prod_{j=1}^d \mu_j$, with μ_j being univariate λ as λ is a measure on $\mathcal{B}(\mathbb{R})$; see E. 2.29. Then $\mu(\prod_{j=1}^d (a_j, b_j]) \underset{\text{def.}}{=} \prod_{j=1}^d \mu_j((a_j, b_j]) = \prod_{j=1}^d (b_j - a_j)$, $-\infty < a_j \leq b_j < \infty$, $j = 1, \ldots, d$, so μ and the d-dimensional λ we constructed coincide on the semiring $\mathcal{A}$, which is a π-system by definition. Also, the d-dimensional λ is

σ-finite as seen in the proof of T. 2.39. By P. 2.28, we thus must have $\mu = \lambda$, so the Lebesgue measure on $\mathbb{R}^d$ is the product measure of the Lebesgue measure on $\mathbb{R}$. We could have thus also shown T. 2.39 for $d = 1$ and defined the d-dimensional Lebesgue measure via the product measure.

Remark 2.40

1) ▪ $\mathcal{A}$ is a semiring but no ring (for $\boldsymbol{a} < \boldsymbol{c} < \boldsymbol{d} < \boldsymbol{b}$, $(\boldsymbol{a}, \boldsymbol{b}] \backslash (\boldsymbol{c}, \boldsymbol{d}]$ is not a hypercube, but a finite disjoint union of hypercubes). To apply Carathéodory's extension theorem for rings, one would consider the ring $\mathcal{A}_{\uplus} := \{\biguplus_{i=1}^n (\boldsymbol{a}_i, \boldsymbol{b}_i] : -\boldsymbol{\infty} < \boldsymbol{a}_i \leq \boldsymbol{b}_i < \boldsymbol{\infty}\ \forall i, n \in \mathbb{N}\}$ generated by $\mathcal{A}$ and show that $\mu_0(\biguplus_{i=1}^n (\boldsymbol{a}_i, \boldsymbol{b}_i]) := \sum_{i=1}^n \Delta_{(\boldsymbol{a}_i, \boldsymbol{b}_i]} F$ is a premeasure on $\mathcal{A}_{\uplus}$; see Steps 2.1)–2.3) in the proof of T. 2.35.

   ▪ $\mathcal{A}_{\uplus}$ is not an algebra (since $\Omega = \mathbb{R}^d \notin \mathcal{A}_{\uplus}$). To apply Carathéodory's extension theorem for algebras, one would consider the algebra $\mathcal{A}' := \{\biguplus_{i=1}^n I_i : I_i = (\boldsymbol{a}_i, \boldsymbol{b}_i] \text{ or } I_i = (\boldsymbol{a}_i, \boldsymbol{\infty}) \text{ for } -\boldsymbol{\infty} \leq \boldsymbol{a}_i \leq \boldsymbol{b}_i < \boldsymbol{\infty}\}$ and show that $\mu_0(\biguplus_{i=1}^n (\boldsymbol{a}_i, \boldsymbol{b}_i]) := \sum_{i=1}^n \Delta_{(\boldsymbol{a}_i, \boldsymbol{b}_i]} F$ is a premeasure on $\mathcal{A}'$.

2) Similarly for left-continuous F and intervals of the form $[\boldsymbol{a}, \boldsymbol{b})$, but less common.

Question: Why does a componentwise increasing F not suffice to construct a measure?

Example 2.41 (Componentwise increasing $\not\Rightarrow$ d-increasing)
Let $F(\boldsymbol{x}) = \max\{(\sum_{j=1}^d x_j) - d + 1, 0\}$, $\boldsymbol{x} \in [0,1]^d$. Then F is componentwise increasing, but not d-increasing for $d \geq 3$ since

$$\begin{aligned}
\Delta_{(\mathbf{1/2},\mathbf{1}]} F &= \sum_{\boldsymbol{i} \in \{0,1\}^d} (-1)^{\sum_{j=1}^d i_j} F((1/2)^{i_1} 1^{1-i_1}, \dots, (1/2)^{i_d} 1^{1-i_d}) \\
&= \max\{1 + \dots + 1 + 1 - d + 1, 0\} \quad (i_j = 0 \; \forall \, j) \\
&\quad - d \max\{1 + \dots + 1 + 1/2 - d + 1, 0\} \quad (\exists! \, j : i_j = 1) \\
&\quad \pm \dots + (-1)^d \max\{d/2 - d + 1, 0\} \quad (i_j = 1 \; \forall \, j) \\
&= 1 - d/2 < 0, \quad d \geq 3,
\end{aligned}$$

so λ_F does not induce a Borel measure on $\mathbb{R}^d$, $d \geq 3$ (since $\lambda_F((\mathbf{1/2}, \mathbf{1}]) < 0$).